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599 Edge AI and Robotics
Module 2 Hello AI World

Objective:

The primary objective of this lab was to configure a Jetson Nano using Docker to deploy and test various AI inference models from NVIDIA's "Hello AI World" project. The lab focused on implementing real-time image classification, semantic segmentation, and pose estimation using RTP and GStreamer.

Setup:

The environment was built by cloning the jetson-inference repository and deploying the project within a Docker container to ensure consistent library dependencies. High level setup steps included:

- 1) **Containerization:** Launched via `docker/run.sh`
- 2) **Remote Visualization:** Configured X11 forwarding via MobaXterm for static images, and used RTP for video streams.
- 3) **Gstreamer Pipeline:** A custom pipeline was used on the host machine to decode the H264 stream sent from the jetson:

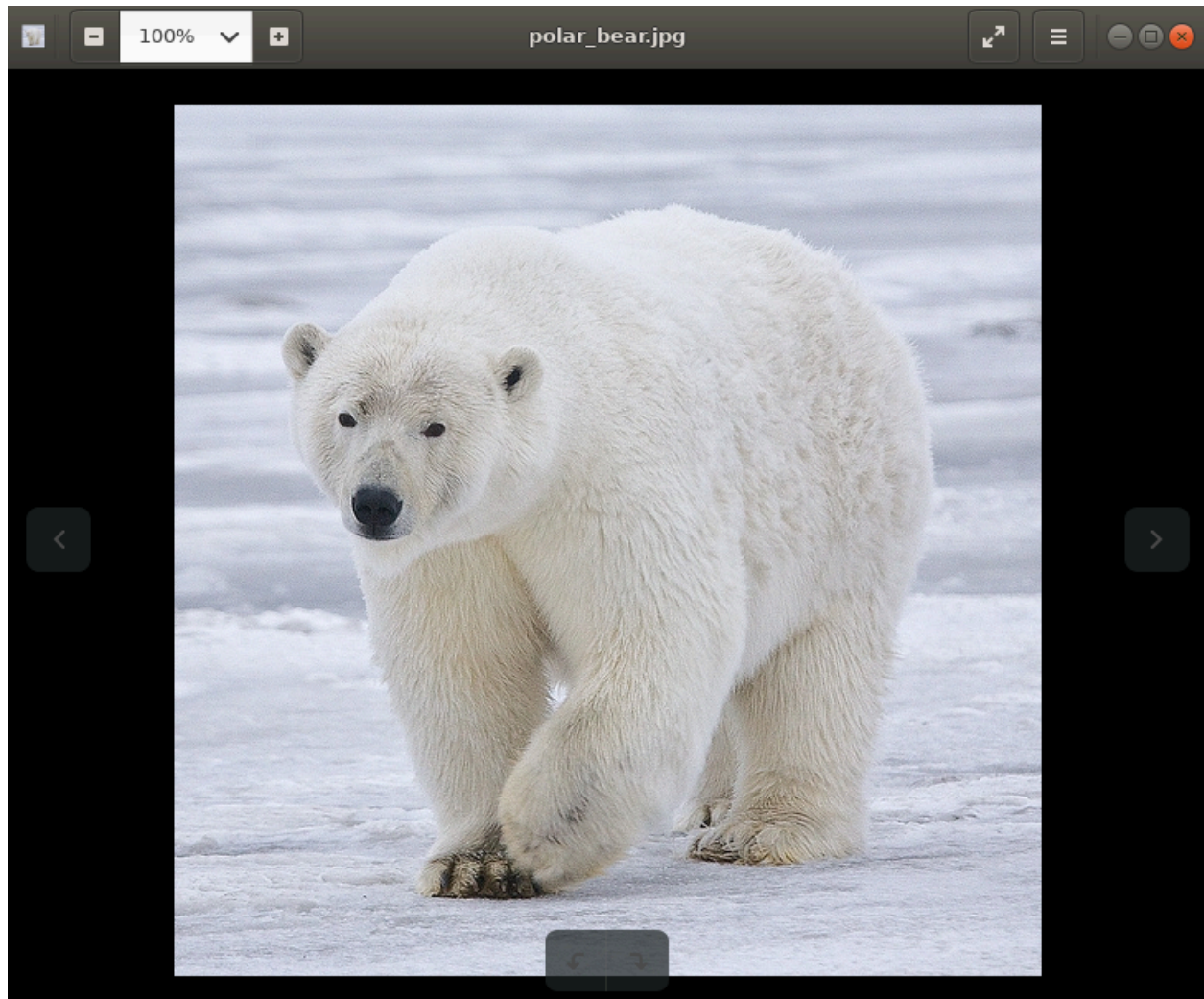
```
`gst-launch-1.0 -v udpsrc port=1234 ! rtph264depay ! decodebin ! autovideosink`
```

Image Classification (ImageNet):

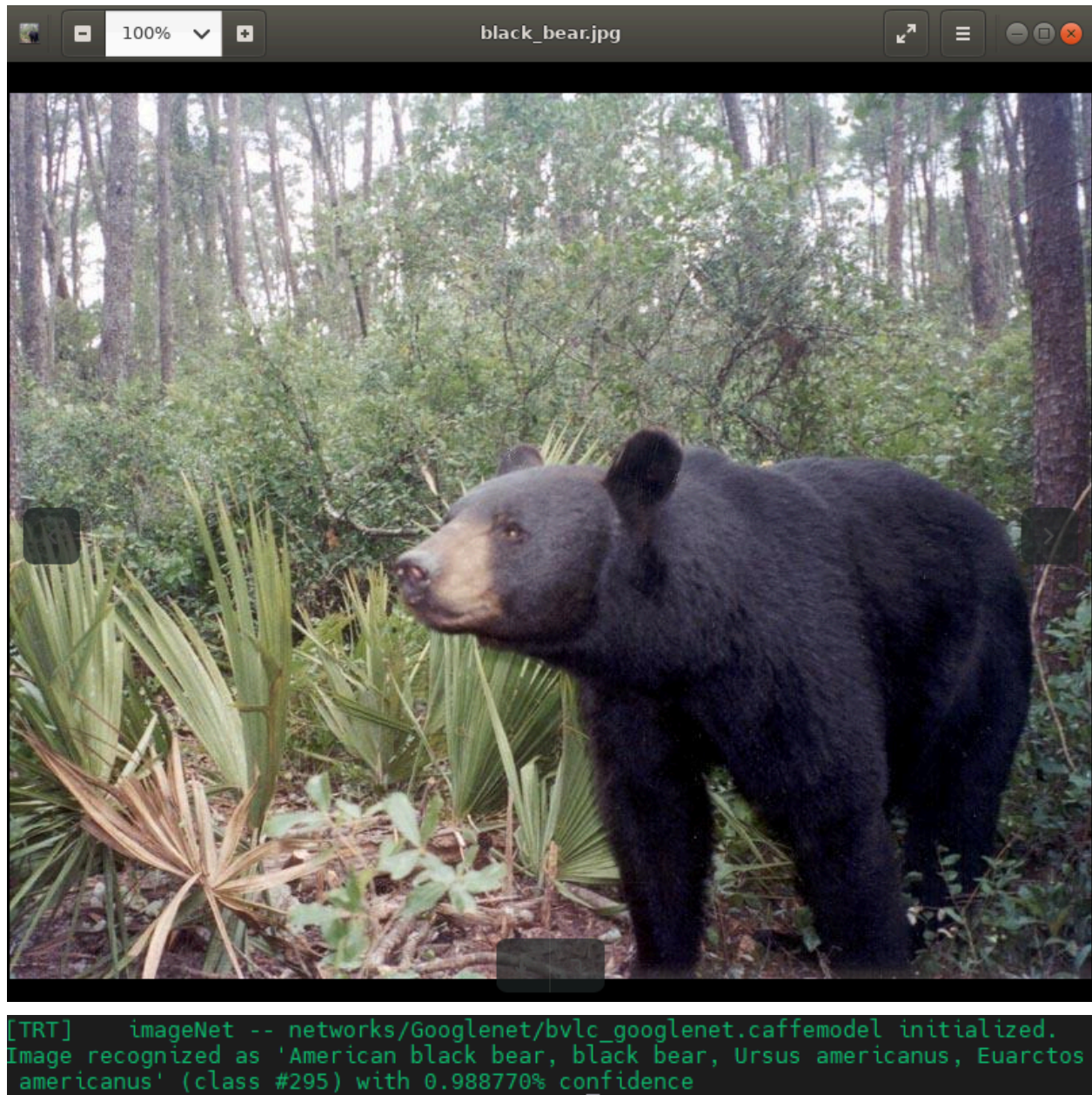
In this exercise, a classification script was used to recognize objects within an image. This process involves the imageNet object, which uses deep learning models such as GoogLeNet or ResNet-18 (you may install additional models) to assign labels and confidence scores to the input data.

Results:

The model was consistent in correctly identifying objects. Remote viewing was performed using the `eog` command to view the image and the predicted class is output by the program.

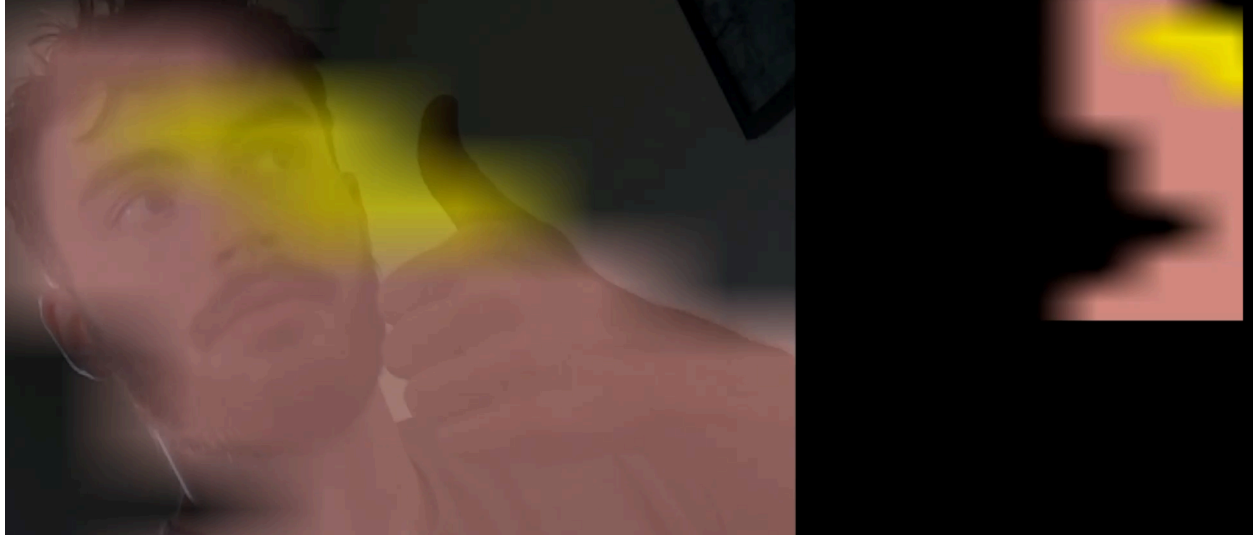


```
[TRT] imageNet -- networks/Googlenet/bvlc_googlenet.caffemodel initialized.  
Image recognized as 'ice bear, polar bear, Ursus Maritimus, Thalarctos maritimus'  
(class #296) with 1.000000% confidence
```



Semantic Segmentation (SegNet):

Unlike simple classification, semantic segmentation classifies every pixel in the image. This is useful for building an environment from an image, such as a self-driving car identifying the driveable portion of the road, or to tell the background apart from the foreground. This was tested on a live video feed and while the performance is limited in terms of fps it does produce an interesting output.

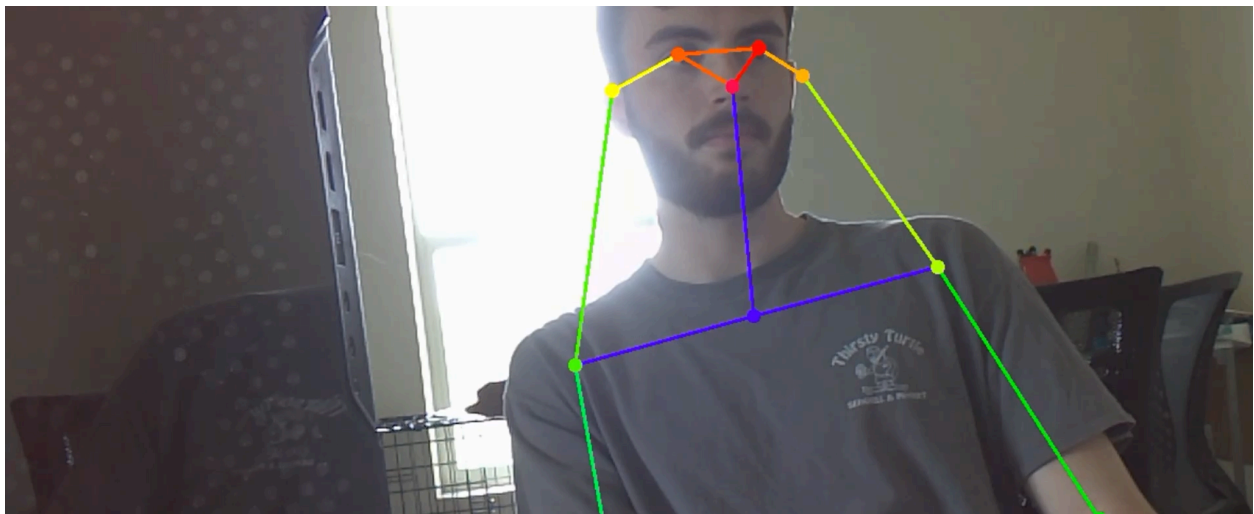


Human Pose Estimation (PoseNet):

The final task involved poseNet, which identifies the coordinates of human body parts (keypoints) to track skeleton movement. This was tested using a live video stream over RTP.

Implementation:

The Jetson Nano processed the video frames, overlaid the skeleton, and transmitted the processed frames over RTP. The RTP stream is then displayed on the host device using the GStreamer CLI



Conclusion:

This lab demonstrated the efficiency of the Jetson Nano in handling complex AI tasks like pixel level segmentation and real time pose tracking. By utilizing Docker and RTP streaming, the limitations of remote development are fairly simple to overcome. It served as a good introduction to a prototyping workflow.